

## QIXIAN SHAN, China

WMO: 587260

Lat: 27.960N

Lon: 117.841E

Elev: 4642

StdP: 12.39

Time Zone: 8.00 (E08)

Period: 82-91

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.7       | 21.6 | -4.9                          | 5.0 | 35.2 | 1.8 | 7.2 | 34.8 | 33.4                  | 40.8 | 28.3 | 40.3 | 5.9                   | 320  | 0.458 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.7                    | 78.7            | 70.6 | 77.0 | 70.0 | 75.5 | 69.1 | 71.9                | 76.7 | 70.9 | 75.6 | 70.0 | 74.4 | 3.7                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.2                            | 132.6 | 75.2 | 69.3 | 128.4 | 74.2 | 68.4 | 124.4 | 73.1 | 39.0          | 77.3 | 38.1 | 75.7 | 37.1 | 74.6 | 79.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 31.7              | 25.5 | 20.8 | 13.7                       | 84.0 | 2.6                | 3.4 | 11.9                                               | 86.5 | 10.3       | 88.5 | 8.9        | 90.4 | 7.0        | 92.9 |  |  |
|                   |      |      | 12.4                       | 74.4 | 2.5                | 2.3 | 10.6                                               | 76.0 | 9.1        | 77.4 | 7.7        | 78.7 | 5.9        | 80.3 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 54.1   | 36.4  | 37.9  | 44.6  | 53.0  | 60.5  | 66.3  | 71.5  | 69.5  | 63.6  | 56.4  | 47.9  | 40.0 |
|                                                                                  | DBStd       | 13.81  | 8.01  | 8.57  | 9.09  | 7.49  | 6.15  | 4.61  | 3.45  | 3.22  | 4.92  | 6.14  | 7.73  | 9.13 |
|                                                                                  | HDD50       | 1507   | 423   | 348   | 216   | 54    | 5     | 0     | 0     | 0     | 0     | 15    | 125   | 321  |
|                                                                                  | HDD65       | 4496   | 885   | 760   | 634   | 362   | 161   | 39    | 4     | 5     | 82    | 275   | 513   | 776  |
|                                                                                  | CDD50       | 2988   | 3     | 8     | 48    | 144   | 330   | 489   | 666   | 605   | 409   | 213   | 63    | 10   |
|                                                                                  | CDD65       | 500    | 0     | 0     | 1     | 2     | 21    | 77    | 205   | 144   | 41    | 8     | 1     | 0    |
|                                                                                  | CDH74       | 680    | 0     | 0     | 2     | 0     | 14    | 64    | 419   | 162   | 17    | 2     | 0     | 0    |
|                                                                                  | CDH80       | 22     | 0     | 0     | 0     | 0     | 1     | 2     | 13    | 5     | 1     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.6    | 7.5   | 8.2   | 8.6   | 9.2   | 8.3   | 8.1   | 8.2   | 6.9   | 6.6   | 6.2   | 6.9   | 6.6  |
| Precipitation                                                                    | PrecAvg     | 76.1   | 3.4   | 4.3   | 8.4   | 9.1   | 11.0  | 14.5  | 6.5   | 7.0   | 3.7   | 2.3   | 3.5   | 2.5  |
|                                                                                  | PrecMax     | 99.4   | 8.8   | 8.9   | 17.4  | 16.5  | 22.0  | 31.6  | 15.5  | 13.0  | 9.8   | 8.3   | 9.8   | 9.3  |
|                                                                                  | PrecMin     | 47.7   | 0.3   | 0.5   | 3.4   | 4.3   | 4.3   | 4.5   | 0.7   | 2.8   | 1.1   | 0.3   | 0.4   | 0.2  |
|                                                                                  | PrecStd     | 15.9   | 1.9   | 2.0   | 3.2   | 3.2   | 4.3   | 6.1   | 3.5   | 3.0   | 1.9   | 2.0   | 2.8   | 2.4  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 56.2  | 59.7  | 66.4  | 71.7  | 75.8  | 78.6  | 81.6  | 79.7  | 76.4  | 73.5  | 68.2  | 58.6 |
|                                                                                  |             | MCWB   | 48.7  | 52.1  | 59.2  | 63.6  | 67.0  | 69.8  | 71.4  | 70.5  | 69.9  | 67.8  | 60.5  | 46.9 |
|                                                                                  | 2%          | DB     | 52.5  | 55.5  | 61.9  | 67.8  | 72.3  | 75.7  | 79.6  | 77.4  | 73.4  | 69.8  | 63.3  | 55.9 |
|                                                                                  |             | MCWB   | 46.4  | 50.6  | 57.3  | 61.5  | 64.7  | 69.2  | 71.1  | 70.2  | 68.0  | 65.2  | 56.9  | 46.4 |
|                                                                                  | 5%          | DB     | 49.9  | 52.7  | 59.2  | 65.4  | 70.3  | 73.9  | 78.0  | 75.9  | 71.5  | 67.0  | 60.2  | 53.3 |
|                                                                                  |             | MCWB   | 45.9  | 49.5  | 55.8  | 59.8  | 64.2  | 68.5  | 70.5  | 69.5  | 66.9  | 62.3  | 54.1  | 45.3 |
|                                                                                  | 10%         | DB     | 47.8  | 50.6  | 57.1  | 62.9  | 68.4  | 72.1  | 76.4  | 74.3  | 69.8  | 64.5  | 57.7  | 51.4 |
|                                                                                  |             | MCWB   | 43.2  | 48.0  | 54.4  | 57.6  | 63.2  | 67.2  | 69.7  | 68.7  | 65.6  | 60.4  | 52.3  | 44.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.0  | 53.6  | 60.1  | 65.4  | 68.4  | 71.9  | 73.7  | 72.7  | 71.0  | 68.8  | 61.9  | 51.7 |
|                                                                                  |             | MCDB   | 54.0  | 56.8  | 65.9  | 70.4  | 72.9  | 75.6  | 79.4  | 77.0  | 74.9  | 72.6  | 66.5  | 54.6 |
|                                                                                  | 2%          | WB     | 48.9  | 51.8  | 58.0  | 63.0  | 67.1  | 70.6  | 72.3  | 71.3  | 69.1  | 65.9  | 59.2  | 49.6 |
|                                                                                  |             | MCDB   | 51.0  | 54.4  | 61.3  | 66.6  | 70.6  | 74.4  | 77.8  | 75.6  | 72.2  | 68.9  | 62.5  | 53.3 |
|                                                                                  | 5%          | WB     | 47.2  | 50.3  | 56.2  | 61.0  | 65.7  | 69.3  | 71.4  | 70.3  | 67.8  | 63.6  | 56.5  | 48.0 |
|                                                                                  |             | MCDB   | 49.1  | 52.4  | 58.8  | 64.2  | 68.9  | 72.9  | 76.7  | 74.4  | 70.5  | 66.2  | 58.9  | 51.8 |
|                                                                                  | 10%         | WB     | 44.9  | 48.5  | 54.5  | 59.2  | 64.5  | 68.1  | 70.3  | 69.4  | 66.5  | 61.6  | 54.4  | 45.9 |
|                                                                                  |             | MCDB   | 46.8  | 50.3  | 56.7  | 62.2  | 67.6  | 71.2  | 75.5  | 73.3  | 68.9  | 63.8  | 56.5  | 49.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.5   | 9.0   | 9.4   | 8.7   | 8.2   | 7.1   | 7.7   | 7.7   | 7.1   | 8.2   | 9.4   | 10.3 |
|                                                                                  |             | MCDBR  | 10.7  | 10.8  | 10.3  | 10.5  | 9.9   | 8.9   | 9.8   | 9.6   | 8.7   | 10.2  | 11.5  | 11.3 |
|                                                                                  |             | MCWBR  | 8.7   | 7.9   | 7.5   | 8.4   | 7.6   | 7.2   | 7.8   | 7.2   | 7.1   | 8.3   | 9.5   | 8.8  |
|                                                                                  | 5% WB       | MCDBR  | 10.4  | 10.0  | 9.7   | 9.2   | 8.7   | 8.0   | 8.8   | 8.5   | 7.6   | 8.6   | 10.1  | 10.4 |
|                                                                                  |             | MCWBR  | 8.3   | 7.9   | 7.4   | 7.4   | 6.8   | 6.6   | 7.9   | 6.9   | 6.5   | 6.8   | 7.8   | 8.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.511  | 0.550 | 0.610 | 0.595 | 0.531 | 0.518 | 0.485 | 0.522 | 0.517 | 0.516 | 0.477 | 0.465 |      |
|                                                                                  | taud        | 1.770  | 1.720 | 1.630 | 1.714 | 1.926 | 1.998 | 2.116 | 1.990 | 1.951 | 1.853 | 1.900 | 1.887 |      |
|                                                                                  | Ebn at Noon | 221    | 226   | 223   | 233   | 248   | 250   | 258   | 248   | 244   | 233   | 231   | 229   |      |
|                                                                                  | Edh at Noon | 61     | 69    | 80    | 76    | 62    | 57    | 50    | 57    | 57    | 60    | 53    | 52    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 688    | 784   | 961   | 1131  | 1262  | 1246  | 1737  | 1535  | 1331  | 1154  | 851   | 774   |      |
|                                                                                  | RadStd      | 147    | 195   | 120   | 161   | 188   | 176   | 163   | 144   | 110   | 186   | 140   | 158   |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (8 neighbors) | +0.54 | N/A     | N/A    | +0.79   | N/A   | N/A   | N/A         | N/A   | +198  | +121  |

Nomenclature: See separate page